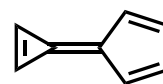
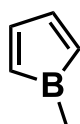
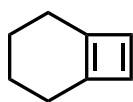
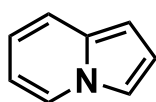
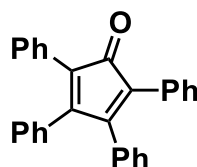
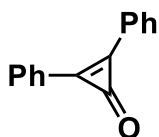


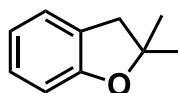
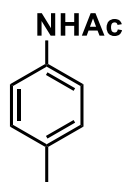
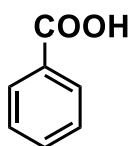
1. Are the following compounds aromatic? Justify your answer.



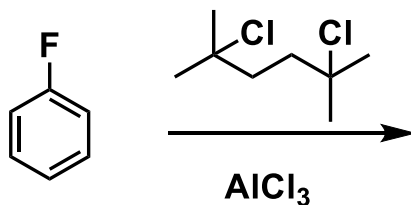
2. Which of the following compounds has the more basic carbonyl oxygen? Explain your answer.



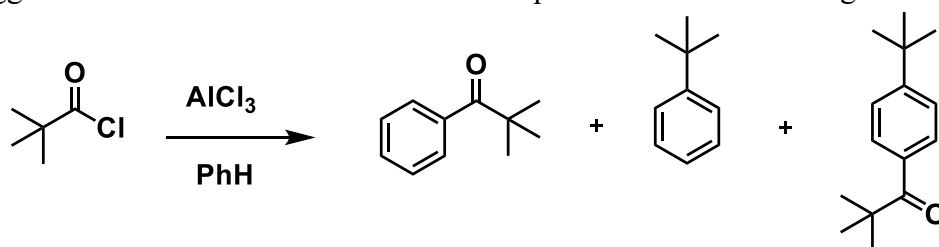
3. Draw the major products of the nitration reactions of the following compounds and suggest a mechanism for the nitration of the last compound.



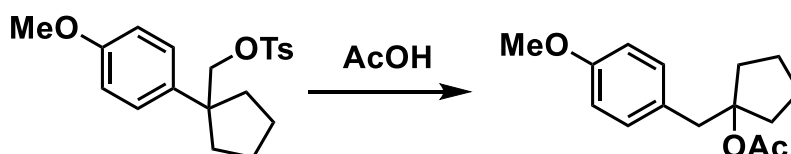
4. Draw the major product for the following reaction and suggest a mechanism.



5. Suggest mechanisms for the formation of all products in the following reaction.

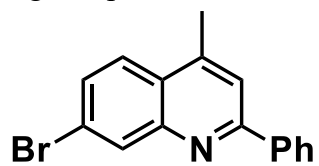


6. Suggest a mechanism for the following reaction.



7. Using organic reagents that contain four carbon atoms or less and necessary inorganic

reagents, synthesize the following compound:



8. Suggest a mechanism for the following reaction.

